

Reconstructing Subsurface Ocean Temperature From Sea Surface Multivariate Remote Sensing Data Using CSSP-ConvLSTM

Jie Sun¹, Jungang Yang², *Member, IEEE*, Wei Cui³, Ole Baltazar Andersen⁴, and Jiawei Xun⁵

Abstract—The 3-D ocean temperature field plays a critical role in studies of global climate change, carbon cycling, and ocean dynamics. Subsurface temperature reconstruction, particularly within the thermocline, remains insufficient due to limitations in observational coverage and modeling frameworks. To address this challenge, we propose the convolutional neural network–Swin Transformer spatial parallel Convolutional Long Short-Term Memory (CSSP-ConvLSTM), a subsurface temperature reconstruction model that integrates multisource sea surface remote sensing data. The model employs a spatial dual-branch structure and a temporal enhancement mechanism to achieve multiscale characterization of local details, global dependencies, and nonlinear evolution. When applied to the northwestern Pacific (30–1000 m), reconstruction accuracy is significantly improved compared with single-branch or single-model baselines, with an average R^2 of 0.9811 and root mean squared error (RMSE) of 0.4557 °C; reconstruction errors are markedly reduced within the thermocline (100–600 m) and deep layers (800–1000 m). Independent evaluations using ARMOR3D and EN4 datasets are conducted, confirming strong temporal generalization, stability, and physical consistency. Interpretability analysis based on SHAP indicates that shallow temperatures are primarily influenced by sea surface temperature (SST) and sea surface height (SSH), whereas subsurface and deep layers are more affected by sea surface salinity (SSS) and SSH. These results demonstrate that the CSSP-ConvLSTM provides an effective approach for reconstructing high-resolution 3-D ocean temperature fields and facilitates improved understanding of thermocline dynamics and their potential climatic implications.

Index Terms—3-D ocean temperature field, deep learning, northwestern Pacific, spatiotemporal reconstruction, subsurface temperature.

I. INTRODUCTION

THE ocean is intricately linked to human society and has a profound influence on global climate change, the carbon cycle, and marine ecosystems [1], [2]. The 3-D ocean

temperature field is a key component of the global thermohaline circulation, capturing the complex pathways of oceanic heat transport and the dynamic evolution of thermodynamic processes [3]. Ocean temperature varies in layers with depth. The surface temperature is higher, the mixed layer temperature is uniform, the thermocline temperature drops rapidly with depth, and the deep layer temperature is relatively constant [4]. The thermocline is a transition zone between the surface and deep layers, characterized by a significant temperature gradient. It is affected by heat flux, wind stress, and vertical mixing, and is of great significance to ocean dynamics and ecological processes [5]. Access to high-precision, high-resolution 3-D ocean temperature fields enables the elucidation of the multiscale processes governing thermohaline structure and mixed-layer evolution, which is critical for improving typhoon intensity forecasts, enhancing El Niño detection, and supporting aquaculture and resource management [6]. Therefore, accurately reconstructing subsurface temperature fields is essential for both fundamental research and practical applications.

Ocean temperature data are typically obtained through satellite remote sensing, in situ observations, and other means [7]. In situ observations are often constrained by high costs, sparse sampling, and limited spatial coverage, making it challenging to resolve basin-wide, high-resolution 3-D temperature fields [8]. Although remote sensing provides extensive spatial coverage, it is restricted to the ocean surface and cannot directly resolve subsurface temperatures [9]. Thus, accurately reconstructing 3-D subsurface temperature fields from surface observations remains a major challenge in ocean science.

Subsurface temperature reconstruction methods have evolved from statistical approaches, dynamical and numerical models, to deep learning methods. Early studies primarily relied on statistical methods [10], [11], [12], dynamical models [13], [14], [15], and surface quasi-geostrophic (SQG) models [16], [17], [18]. These methods yielded reasonable reconstruction results under specific regional or seasonal conditions. However, for high-resolution subsurface temperature reconstruction across regions and seasons, such methods are often sensitive to the underlying dynamical background, parameterization schemes, and initial/boundary conditions, which constrain their applicability to rapid and fine-scale reconstructions driven by multisource observations. In recent years, the rapid development of deep learning has opened new directions for marine science research. With its excellent nonlinear mapping and multilayer representation capabilities, deep learning methods have demonstrated signifi-

Received 3 November 2025; revised 5 January 2026; accepted 25 January 2026. Date of publication 3 February 2026; date of current version 20 February 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62231028 and in part by the European Space Agency-Ministry of Science and Technology of the People's Republic of China (ESA-MOST) Cooperation Dragon 6 Project under Grant 95373. (Corresponding author: Jungang Yang.)

Jie Sun is with the College of Intelligent Systems Science and Engineering, Harbin Engineering University, Harbin 150001, China (e-mail: sunjie_sonia@163.com).

Jungang Yang and Wei Cui are with the Laboratory of Marine Physics and Remote Sensing, First Institute of Oceanography, Ministry of Natural Resources, Qingdao 266061, China (e-mail: yangjg@fio.org.cn; cuiwei@fio.org.cn).

Ole Baltazar Andersen is with DTU Space, Technical University of Denmark, 2800 Kongens Lyngby, Denmark (e-mail: oa@space.dtu.dk).

Jiawei Xun is with the College of Oceanography and Space Informatics, China University of Petroleum (East China), Qingdao 266580, China (e-mail: Z23160098@s.upc.edu.cn).

Digital Object Identifier 10.1109/TGRS.2026.3660826

cant advantages in tasks such as ocean subsurface temperature reconstruction, prediction, and numerical modeling [19], [20], [21]. The convolutional neural network (CNN), which captures the spatial information of data to extract local spatial features, has become a key tool for characterizing ocean spatial structure [22], [23]. Experiments with Argo temperature and salinity data demonstrate that CNN significantly improves reconstruction accuracy within the 0–1000 m depth range, particularly enhancing the resolution of mesoscale signals [24]. With the continuous improvement of model structure, the dual-path CNN (DP-CNN) has further improved reconstruction accuracy [25]. In addition, the convolutional block attention module-CNNs (CBAM-CNNs) have demonstrated superior 3-D temperature reconstruction capabilities when integrating satellite remote sensing with multisource observations, such as Argo [26]. At the same time, the advent of the Transformer architecture, especially the shifted window Transformer (Swin Transformer), effectively models local and long-range dependencies through hierarchical attention, providing new solutions for the representation of complex spatial structures [27], [28].

The ocean subsurface temperature is not solely determined by spatial structural features. Its evolution is shaped by mechanisms, such as heat flux transport, wind stress, and vertical mixing, and exhibits pronounced nonlinearity and temporal dependence. In the thermocline, the temperature field exhibits strong vertical gradients, multiscale spatial variability, and evident temporal lags; as a result, models that rely on either spatial-only or temporal-only representations are unlikely to adequately characterize the coupled spatiotemporal evolution. Long short-term memory (LSTM) networks have excellent performance in capturing temporal dependencies and have been widely applied to ocean temperature prediction [29], [30], [31]. Convolutional LSTM (ConvLSTM) networks further capture temporal dynamics while preserving spatial structure, making them a key tool for spatiotemporal coupling [32]. Multilayer ConvLSTM models have achieved promising results in sea surface temperature (SST) prediction [33], and the incorporation of shallow (such as 50 m) reconstructions to guide deeper layers (100–500 m) has further improved prediction accuracy and stability [34].

Ocean physical variability is characterized by multiscale processes, encompassing large-scale circulation, mesoscale eddies, small-scale fronts, and microscale turbulence, and displays periodic variability from seasonal to decadal scales [35]. Therefore, a spatial dual-branch spatiotemporal architecture is proposed, termed CSSP-ConvLSTM, integrating CNN, Swin Transformer, and ConvLSTM. The architecture is designed to extract multidimensional deep features from multisource sea surface remote sensing data, thereby capturing thermocline evolution more effectively and reconstructing subsurface ocean temperature fields. The main innovations of this study are as follows.

1) For spatial feature extraction, a CNN–Swin Transformer dual-branch design is employed, combining CNN local receptive fields with Swin Transformer hierarchical attention to enhance multiscale spatial representation. This design is particularly targeted at the thermocline (100–600 m), enabling more

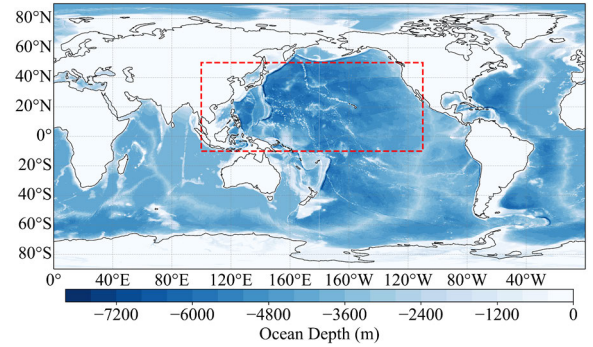


Fig. 1. Location of the study area. The sea area within the red dashed box is the test area.

effective resolution of complex vertical gradients and improving the accuracy of thermocline temperature reconstruction.

2) For temporal feature extraction, ConvLSTM is integrated with temporal channel attention (TCA) and squeeze and excitation (SE) modules to capture temporal dependencies. These modules enhance salient features and suppress redundant information, thereby improving dynamic reconstruction capability in the thermocline and mid-depth layers.

3) A shared feature-extraction module wrapped in TimeDistributed is adopted to perform independent spatial encoding at each time step, ensuring complete spatial representation while preventing loss of temporal information.

This article is organized as follows. Section II introduces the study area and datasets. Section III describes the CSSP-ConvLSTM model, covering its architecture, modules, configuration, evaluation, and parameter tuning. Section IV presents the results and discussion, including model comparisons, spatiotemporal and vertical analyses, accuracy assessment with the EN4 data, and SHAP-based interpretation. Section V concludes this study.

II. STUDY AREA AND DATASETS

A. Study Area Selection

The study area encompasses the northwestern Pacific (10°S–50°N and 100°E–110°W) (see Fig. 1). The ocean temperature in this area is influenced by monsoon forcing, the warm current system, and complex bathymetry, resulting in pronounced horizontal and vertical temperature gradients. In particular, the thermocline is well defined and exhibits strong seasonal variability (approximately 100–300 m in summer and 100–600 m in winter) [36], playing a key role in regulating regional heat exchange and dynamical processes. However, observations of the thermocline and subsurface temperatures in this area remain limited. Therefore, reconstructing subsurface temperatures in this area not only facilitates a comprehensive depiction of the 3-D ocean temperature field but is also of significance for ocean dynamics research, climate prediction, disaster prediction, and early warning.

B. Dataset Composition

The data used in this study are drawn from multi-source ocean surface remote sensing observations and 3-D data products. Ocean surface remote sensing observations

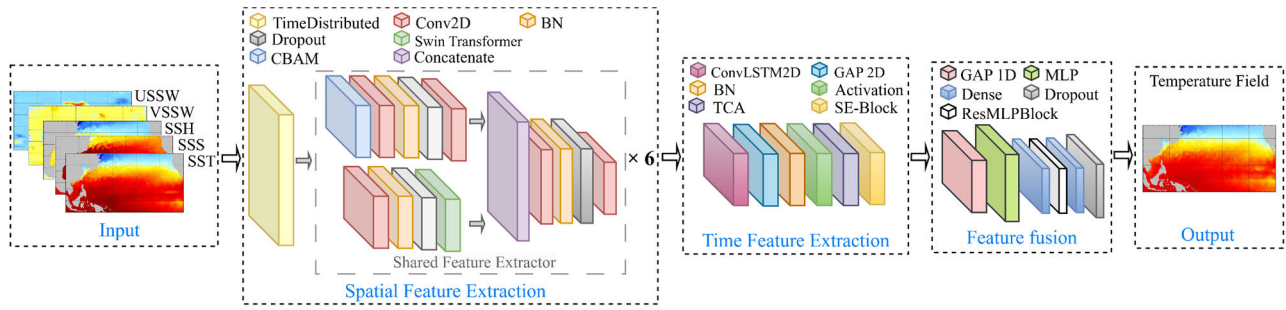


Fig. 2. Schematic of the model architecture.

include SST, sea surface salinity (SSS), sea surface height (SSH), and sea surface wind speed (SSW). SST data are provided by U.K. Meteorological Office [(UKMO), <https://www.metoffice.gov.uk/>] and are generated from a combination of multisatellite and in situ buoy observations, with a daily temporal resolution and a spatial resolution of $0.05^\circ \times 0.05^\circ$ [37]. Both SSH and SSS are provided by the Copernicus Marine Environment Monitoring Service [(CMEMS), <https://marine.copernicus.eu/>]. SSH data are obtained from a combination of multisatellite observations, with a daily temporal resolution and a spatial resolution of $0.25^\circ \times 0.25^\circ$ [38]. SSS data are obtained by the fusion of multisource satellite and in situ observations, with a monthly temporal resolution and a spatial resolution of $0.125^\circ \times 0.125^\circ$ [39]. The SSW data are from the Cross-Calibration Multi-Platform [(CCMP), <https://podaac.jpl.nasa.gov/MEASURES-CCMP>] project and contain zonal (USSW) and meridional (VSSW) components, with a monthly temporal resolution and a spatial resolution of $0.25^\circ \times 0.25^\circ$ [40]. The 3-D data products used are the Multi Observation Global Ocean ARMOR3D L4 analysis and multi-year reprocessing (hereafter ARMOR3D), the Copernicus Global $1^\circ/12^\circ$ Oceanic and sea ice physical reanalysis (GLORYS12V1) ocean reanalysis, and the quality-controlled global ocean subsurface temperature and salinity dataset (EN4). ARMOR3D data are provided by CMEMS and are available on 50 standard depth layers, with weekly or monthly temporal resolution and a spatial resolution of $0.25^\circ \times 0.25^\circ$ [41]. GLORYS12V1 is a global ocean reanalysis product provided by CMEMS, generated by assimilating multisource observations into a numerical ocean model, with 50 vertical layers, daily temporal resolution, and a spatial resolution of $1^\circ/12^\circ \times 1^\circ/12^\circ$ [42]. The independent model evaluation is based on the EN4 data provided by the U.K. Met Office (EN4.2.2, <https://www.metoffice.gov.uk/hadobs/en4/>), spanning 83°S – 90°N , comprising 42 standard depths, with a monthly temporal resolution and a spatial resolution of $1^\circ \times 1^\circ$ [43].

III. METHODOLOGY

To fully integrate the spatial and temporal characteristics of multitime series ocean surface observations and accurately reconstruct the ocean subsurface temperature field, a spatiotemporal feature extraction model, integrating CNN, Swin Transformer, and ConvLSTM, is developed. The model is built upon an enhanced joint channel–temporal attention mechanism, which effectively extracts the temporal evolution

and spatial distribution features from multisource sea surface observations, and reconstructs subsurface temperature fields at 15 designated depth layers (30, 50, 100, 125, 150, 200, 250, 300, 400, 500, 600, 700, 800, 900, and 1000 m). The model architecture is shown in Fig. 2. The model consists of three main modules: the dual-branch spatial feature extraction module, the temporal feature extraction module, and the spatiotemporal feature fusion module. The dual-branch spatial feature extraction module employs CNN and Swin Transformer branches to capture spatial features at different scales. The temporal feature extraction module employs an improved ConvLSTM to extract representative spatiotemporal features. Finally, the spatiotemporal feature fusion module integrates the extracted information to produce the reconstructed ocean temperature.

Taking into account both model accuracy and computational resources, the model input is defined as a six-month multivariate ocean surface remote sensing tensor with dimensions (6, 9, 9, 5), corresponding to six time steps, a 9×9 spatial grid, and five channels (SSH, SSS, SST, USSW, and VSSW). The six-month consecutive ocean surface observation sequences are processed using a sliding time window, whereby the ocean surface observations from the preceding six months are used to reconstruct the subsurface temperature field of the current month.

A. Dual-Branch Spatial Feature Extraction Module

To achieve independent spatial encoding at each time step, a shared feature extractor module wrapped in TimeDistributed is employed to independently extract spatial features. In order to fully explore the multiscale spatial information contained in ocean surface observation data and enhance the ability of the model to recognize features of different scales, a dual-branch spatial feature extraction module integrating the Swin Transformer and CNN is designed. In addition, channel and spatial attention mechanisms from the convolutional block attention module (CBAM) are incorporated to enhance the responsiveness of the model to key regions. The architecture of this module is illustrated in Fig. 3. Finally, the outputs of the two branches are dimensionally reduced and fused through an integration mechanism and are subsequently fed into the ConvLSTM for temporal feature extraction.

1) *CNN Branch*: To extract local spatial edges and structural details, a parallel lightweight CNN branch is designed. This branch comprises a multilayer 2-D CNN, combined with rectified linear unit (ReLU) activation and batch normalization

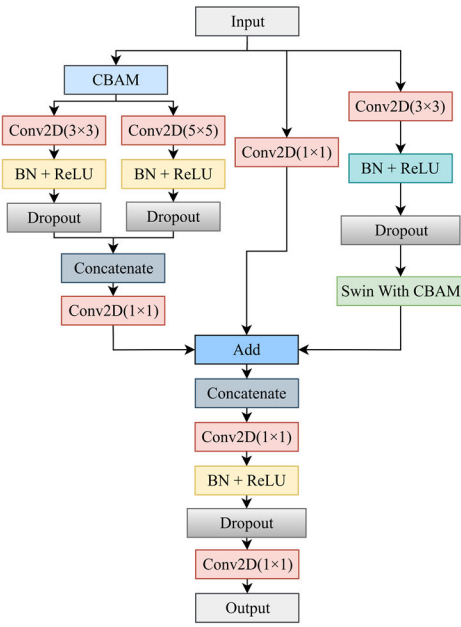


Fig. 3. Schematic of the spatial feature extraction module.

(BN), to facilitate stable and efficient encoding of spatial information. The basic convolution operations consist of combinations of 3×3 , 5×5 , and 1×1 convolutional filters, enabling the capture of complex spatial variation patterns at multiple scales. These operations are used to extract the local spatial features of each variable. Then, CBAM is applied after the convolutional outputs. CBAM consists of channel and spatial attentions. Channel attention is used to identify key variable channels through global pooling and fully connected operations, while the latter highlights important spatial regions through maximum pooling and convolution operations [44].

2) *Swin Transformer Branch*: The Swin Transformer architecture is centered on the shifted window attention mechanism, which is designed to effectively reduce computational complexity while capturing long-range spatial dependencies within local regions. The central idea is that self-attention operations are performed within fixed-size windows, while cross-region interactions are enhanced through the shifted window attention mechanism.

In this branch, the input data are first partitioned into local windows, within which multihead self-attention mechanisms are applied to fully extract local spatial dependencies. To alleviate the problem of information isolation between windows, the shifted window attention mechanism is introduced to enable information exchange across windows, thereby enhancing the global representational capacity while maintaining computational efficiency. To enhance the global spatial structure and the capability of the model to extract locally significant features, the outputs of this branch are subsequently passed through the CBAM module, which strengthens the responsiveness of key variable dimensions and spatial dimensions. This architecture can effectively capture the influence of local structures on subsurface temperature.

3) *Dual-Branch Integration Module*: The outputs of the Swin Transformer and CNN branches are integrated through feature concatenation. To reduce model complexity and com-

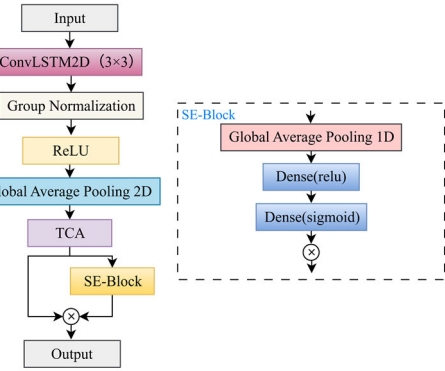


Fig. 4. Schematic of the temporal feature extraction module structure.

putational burden, 1×1 convolution and global average pooling (GAP) [45] operations are employed to compress high-dimensional spatial features into fixed-length spatial representation vectors. Specifically, 1×1 convolution is used for dimensionality reduction and feature fusion, which not only reduces the number of channels but also enables lightweight feature interactions; GAP is applied to transform the 2-D spatial structure into a 1-D representation, thereby facilitating the subsequent fusion and regression analysis of temporal information.

B. Temporal Feature Extraction Module

In consideration of the continuous evolution of ocean dynamics, the temporal feature extraction module is constructed on an improved ConvLSTM, further combined with TCA and the SE-Block [46], [47] to capture temporal dependencies and trends. The architecture of the module is illustrated in Fig. 4.

ConvLSTM is capable of modeling both spatial structures and temporal evolution, and is thus suitable for processing spatial sequence data. To mitigate accumulated temporal errors and improve gradient stability, layer normalization (LN) is applied after the ConvLSTM output to standardize activation distributions at each time step, thereby enhancing the stability and convergence of temporal learning. A lightweight TCA module is subsequently employed to enhance the sensitivity to temporal dependencies within the sequence information. TCA first aggregates channel features within each time step into a temporal sequence and subsequently assigns weights to key time steps through a two-layer fully connected network. This mechanism focuses on representative historical moments and enhances their significance in subsequent use, thereby providing a robust temporal modeling capability. To further strengthen the representational capacity of key channels, the SE-Block is incorporated, which recalibrates each channel using global contextual information, thereby enhancing the response of effective features while suppressing redundant information. Finally, a GAP 1-D layer is applied to compress the temporal features into a fixed-length representation, and the temporal features are then fed into the subsequent fully connected regression layer to reconstruct the subsurface temperature.

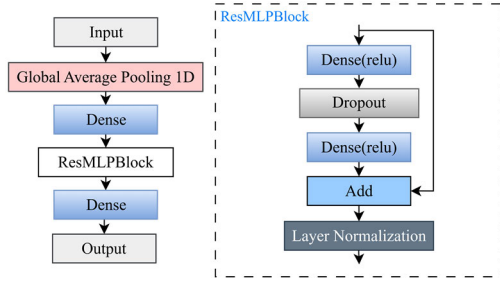


Fig. 5. Schematic of the spatiotemporal feature fusion module structure.

C. Spatiotemporal Feature Fusion Module

After the spatial and temporal features are extracted, the residual MLP block (ResMLP-Block) is employed to enhance feature integration and maintain stability. This module consists of two pointwise fully connected layers, introduces dropout to mitigate the risk of overfitting, and employs residual connections together with LN to achieve robust cross-channel feature fusion. The fused features are fed into a fully connected layer, with $L2$ regularization applied to mitigate overfitting due to over-parameterization, and dropout applied to improve generalization performance. Finally, the temperature reconstruction result is generated through a linear output layer. The module architecture is illustrated in Fig. 5. This structure effectively enhances the representational capacity and training convergence of the model while maintaining a relatively small number of parameters, making it suitable for scenarios involving complex multivariable-to-univariable mappings.

D. Model Configuration

The monthly mean subsurface temperature field is reconstructed from SSH, SSS, SST, USSW, and VSSW data as inputs. The raw SST, SSS, and SSH data are first spatially and temporally resampled using bilinear interpolation and averaging, yielding a unified resolution of monthly means at $0.25^\circ \times 0.25^\circ$. The data are then normalized. Furthermore, to ensure that each grid point contains complete surface observations and subsurface temperature labels, grid points with missing values are excluded. Finally, a total of 156 gridded monthly mean data are obtained from January 2010 to December 2022. Data from 2011 to 2021 are used as the training set, with 70% randomly selected for model training and 30% for model validation. Data from 2010 and 2022 are used as the test set to evaluate the accuracy and generalization ability of the model.

During the training process, the masked mean squared error (Masked mse) loss function is employed, which computes errors only at locations with valid ground-truth values, thereby avoiding the influence of invalid values on model training. The Adam optimizer with an adaptive learning rate is adopted and coupled with the mixed-precision training strategy with TensorFlow to accelerate training while conserving GPU memory. The initial learning rate is set to 0.001, and a ReduceLROn-Plateau strategy [48] is employed for dynamic adjustment. When the loss does not improve over eight consecutive epochs, the learning rate is automatically reduced. To prevent the

TABLE I
RECONSTRUCTION ACCURACY OF DIFFERENT
BATCH SIZES

Batch size	R^2	RMSE ($^\circ\text{C}$)
512	0.9789	0.4559
1024	0.9795	0.4561
2048	0.9802	0.4566
4096	0.9811	0.4557
5120	0.9812	0.4558

TABLE II
ABLATION EXPERIMENTS OF DIFFERENT
MODEL STRUCTURES

Case	Model	R^2	RMSE ($^\circ\text{C}$)
Case1	CNN-ConvLSTM	0.9786	0.4734
Case2	Swin-ConvLSTM	0.9783	0.4771
Case3	CSSP-ConvLSTM	0.9811	0.4557
Case4	CSSP-ConvLSTM without TCA	0.9803	0.4671
Case5	CSSP-ConvLSTM without SE-Block	0.9799	0.4678
Case6	CSSP-ConvLSTM without TimeDistributed	0.9806	0.4665

Note: All models are trained and evaluated under identical data splits and training settings to ensure a fair comparison. The reported R^2 and RMSE values are averaged over all depths in the test period. The “without” denotes removing the corresponding module from the full CSSP-ConvLSTM architecture.

model from overfitting in the later stages of training, an early stopping strategy is adopted. When the validation set loss does not decrease significantly within 15 consecutive epochs, training is automatically stopped, and the best weights are restored. The maximum training is set to 200 epochs.

E. Model Performance Indicators

In order to comprehensively evaluate the reconstruction ability of the model, the root mean squared error (RMSE) and the coefficient of determination (R^2) are employed as metrics of model performance. The specific definition is as follows:

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^n (x_{\text{real},i} - x_{\text{inverse},i})^2}{n}} \quad (1)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (x_{\text{real},i} - x_{\text{inverse},i})^2}{\sum_{i=1}^n (x_{\text{real},i} - \bar{x}_{\text{real}})^2} \quad (2)$$

where n is the number of data points, $x_{\text{real},i}$ and $x_{\text{inverse},i}$ are the observations and reconstructions at position i , respectively, and $\bar{x}_{\text{real}}$ is the mean of the observations.

F. Model Parameter Tuning

The subsurface ocean temperature field for 2022 is reconstructed to analyze the impact of batch size settings on reconstruction accuracy and to determine the optimal configuration. The results indicate that as the batch size increases, the R^2 of the model tends to increase overall, while the RMSE ranges between 0.4557°C and 0.4566°C , as shown in Table I. It can be observed that when the batch size is 4096 and 5120,

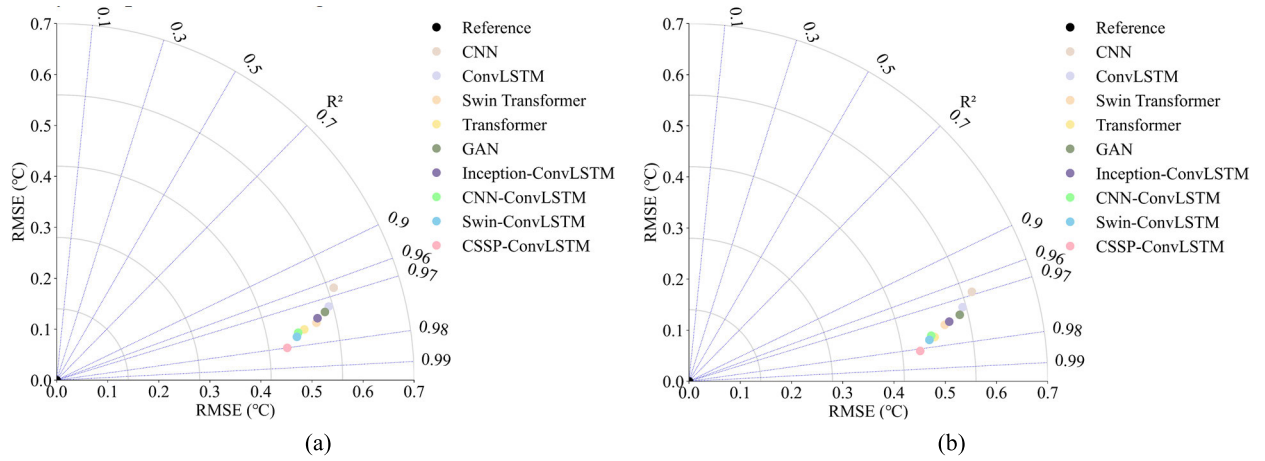


Fig. 6. Taylor diagram comparing the performance of different models. Statistics are computed over all grid points and all depth layers in the test year (a) 2010 and (b) 2022.

the R^2 reaches 0.9811 and 0.9812, respectively, which are significantly higher compared with smaller batch sizes, while the corresponding RMSE remains at a relatively low level (0.4557 °C). Therefore, a batch size of 4096 is adopted to ensure model accuracy while maintaining a balance between training efficiency and stability.

In addition, ablation experiments are conducted to further assess the contributions of the spatial dual-branch backbone and the key components, including TCA, SE-Block, and the TimeDistributed encoder (see Table II). All six model variants achieve strong reconstruction skill, while the full CSSP-ConvLSTM consistently exhibits the best performance. Compared with the single-branch baselines (CNN-ConvLSTM and Swin-ConvLSTM) under otherwise identical settings, CSSP-ConvLSTM exhibits a clear improvement in reconstruction accuracy, indicating that the dual-branch design better captures both local spatial details and nonlocal spatial dependencies, which is beneficial for subsurface temperature reconstruction. Furthermore, removing either TCA or SE-Block leads to noticeable performance degradation, suggesting that TCA and channel recalibration play important roles in representing temporal evolution and stratification-related signals. Removing the TimeDistributed module results in a smaller yet nonnegligible decrease in performance, indicating that shared per-time-step encoding contributes to improved temporal consistency.

IV. RESULTS AND DISCUSSION

A. Quantitative Analysis

To evaluate the reconstruction performance of CSSP-ConvLSTM, several representative baseline models are considered, including CNN, ConvLSTM, Swin Transformer, a temporal-spatial Transformer, GAN, multiscale Inception-ConvLSTM, CNN-ConvLSTM, and Swin-ConvLSTM. Comparative experiments were conducted in the study area to assess their performance in reconstructing the subsurface temperature field.

Fig. 6 presents the domain-averaged reconstruction performance for 2010 and 2022. It is evident from Fig. 6 that

the CSSP-ConvLSTM achieves the best performance in both years, with R^2 values of 0.9809 and 0.9811, and RMSE values of 0.4555 °C and 0.4557 °C, all superior to those of the other models. This consistency across years indicates strong temporal generalization and stability.

Fig. 7 shows the depth-resolved reconstruction performance of different models for 2010 and 2022. CSSP-ConvLSTM maintains high R^2 and low RMSE across all depth layers in both years, with particularly pronounced advantages in the thermocline (100–600 m) and deep layers (800–1000 m). These results indicate that the model effectively captures the strong vertical temperature gradients characteristic of the thermocline while preserving stable reconstruction skill in deep layers with weaker variability. In contrast, CNN-ConvLSTM and Inception-ConvLSTM, which primarily rely on local receptive fields, show reduced reconstruction skill in regions where large-scale spatial context is essential, leading to lower accuracy in the thermocline. Swin-ConvLSTM benefits from attention-based spatial modeling and exhibits improved performance in the thermocline; however, its reconstruction capability in deeper layers remains limited. This suggests that a single spatial modeling branch is insufficient to simultaneously represent multiscale spatial structures and their temporal evolution. The temporal-spatial Transformer outperforms conventional CNN, ConvLSTM, and baseline GAN models in overall accuracy; however, its performance remains inferior to CSSP-ConvLSTM, likely due to the absence of an explicit recurrent memory mechanism for sequential dynamics. By integrating dual-branch CNN and Swin Transformer spatial encoders with ConvLSTM-based temporal modeling, CSSP-ConvLSTM achieves effective coordination between multiscale spatial features and temporal evolution. As a result, the proposed model demonstrates superior reconstruction performance across different years and depth ranges, indicating good temporal generalization and robustness.

To further quantify the statistical reliability of the reconstruction results, the mean error and the associated 95% confidence interval (CI) are analyzed for 2010 and 2022 [49], [50], as shown in Fig. 8. Across the full depth range, the mean error remains largely within -0.1 °C to 0.1 °C,

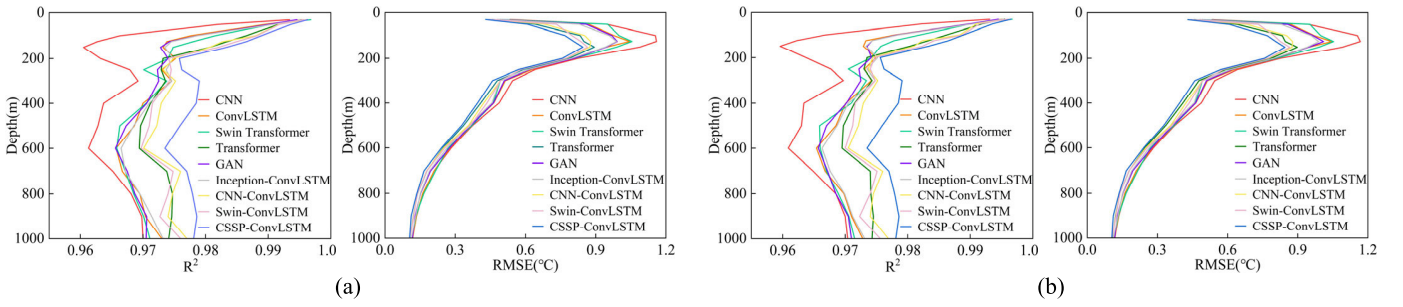


Fig. 7. Performance comparison of different models at different depth layers. R^2 and RMSE are the spatial statistical averages over all 12 months of the year at that depth layer (a) 2010 and (b) 2022.

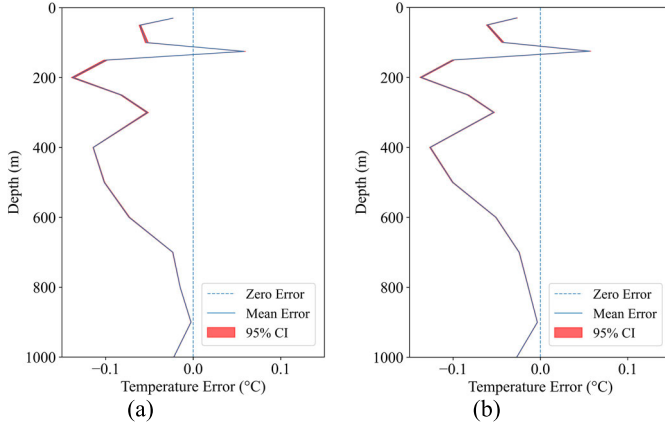


Fig. 8. Uncertainty analysis of reconstruction results across depth layers. The red shaded area denotes the 95% CI estimated from spatial samples, and its width reflects the uncertainty of the spatially averaged reconstruction error at each depth. This analysis characterizes statistical uncertainty in a spatial-mean sense, rather than pointwise reconstruction errors or intrinsic model uncertainty (a) 2010 and (b) 2022.

indicating the absence of pronounced systematic bias. The 95% CI is generally narrow for most depth levels, suggesting that the spatially averaged reconstruction error is estimated with relatively high statistical stability. A modest widening of the CI is observed between 50 and 300 m, which is consistent with the stronger vertical temperature gradients and more complex dynamical processes in and around the thermocline, leading to increased uncertainty in the estimation of mean reconstruction error. Overall, the results indicate that CSSP-ConvLSTM exhibits stable reconstruction performance across different years and depth levels, and that the reconstructed subsurface temperature fields are statistically reliable.

Given that no significant differences in reconstruction accuracy are observed between 2010 and 2022 across the evaluated models, the results suggest that the model exhibits good temporal generalization. Accordingly, 2022 is selected as a representative year for subsequent analyses focusing on depth-dependent reconstruction performance.

B. Spatiotemporal Distribution Characteristics

As a case study, the reconstruction results at different depths (30, 150, 500, and 1000 m) in representative months of different seasons in 2022 (March, June, September, and December) are examined by comparing the differences between reconstructed temperatures from different models with ARMOR3D temperatures.

Fig. 9 shows the spatial distribution of temperature differences between reconstructed temperatures from different models and ARMOR3D at a representative thermocline depth of 150 m. The results of different months indicate that the CSSP-ConvLSTM model more accurately captures the temperature structure in regions with strong variability, such as the Kuroshio Extension and the vicinity of the equatorial circulation, and the deviations between the reconstructed temperatures and ARMOR3D are markedly reduced. Overall, the CSSP-ConvLSTM model achieves the highest reconstruction accuracy in the thermocline, followed by the Swin-ConvLSTM model, while the CNN-ConvLSTM model shows the lowest accuracy. This result is consistent with the conclusions drawn from the comparison of evaluation metrics in Section IV-A.

The differences between the CSSP-ConvLSTM reconstructed temperatures and the ARMOR3D temperatures at different months and depths are further analyzed, as shown in Fig. 10. The results indicate that the model maintains high accuracy at 30, 500, and 1000 m depths, although pronounced temporal and spatial variability exists across these depths. At 30 m, which lies within the surface mixed layer, the overall reconstruction error remains small but shows clear seasonal fluctuations, with accuracy decreasing notably in June and September [see Fig. 10(a)]. This phenomenon is closely related to the frequent typhoon activities during May–October [51]. Substantial perturbations of seawater temperature in the 0–100 m layer caused by typhoons lead to increased reconstruction errors in shallow layers [52]. A 500 m depth lies in the lower part of the thermocline, characterized by a large vertical temperature gradient and strong spatiotemporal variability, yet the model effectively reproduces its main thermal structure [see Fig. 10(b)]. A 1000 m depth belongs to the deep ocean, where the temperature field is stable and only weakly influenced by surface processes [see Fig. 10(c)]. The 500 and 1000 m depths do not exhibit significant seasonal characteristics. The differences between the model and ARMOR3D range from -0.5°C to 0.5°C , with a uniform spatial distribution and no systematic deviations. This indicates the stability and reliability of the model in reconstructing deep-layer temperature.

Overall, the CSSP-ConvLSTM model achieves high-accuracy reconstruction of ocean temperature at all depths throughout the year. The deep layers (500 and 1000 m) exhibit minimal error and uniform distribution, demonstrating excellent stability. Although the shallow layer (30 m) experiences some fluctuations under the influence of seasonal disturbances,

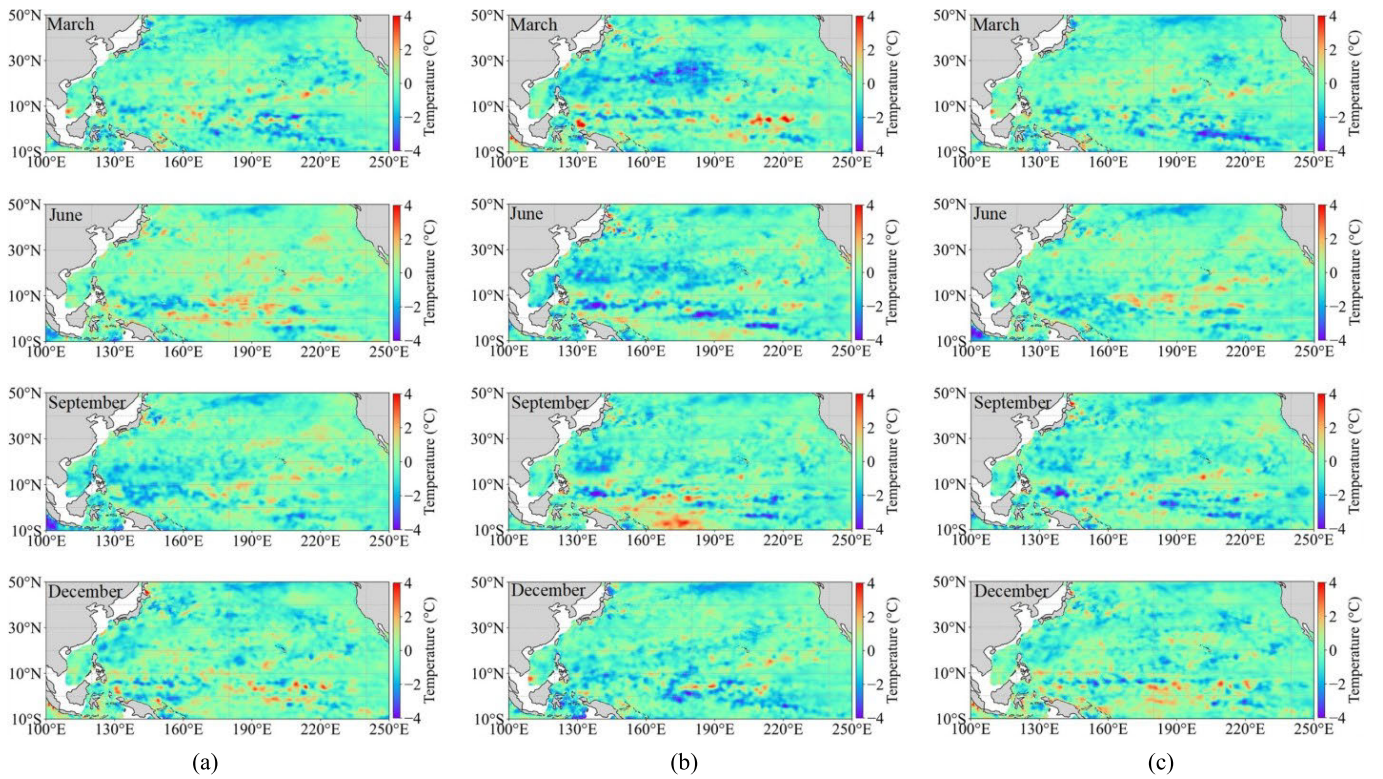


Fig. 9. Difference between the reconstructed temperatures from different models and the ARMOR3D temperatures in different seasons and models at 150 m. Difference denotes the signed temperature difference between reconstructed temperatures and ARMOR3D temperatures, with positive (negative) values indicating warm (cold) bias. (a) CSSP-ConvLSTM. (b) CNN-ConvLSTM. (c) Swin-ConvLSTM.

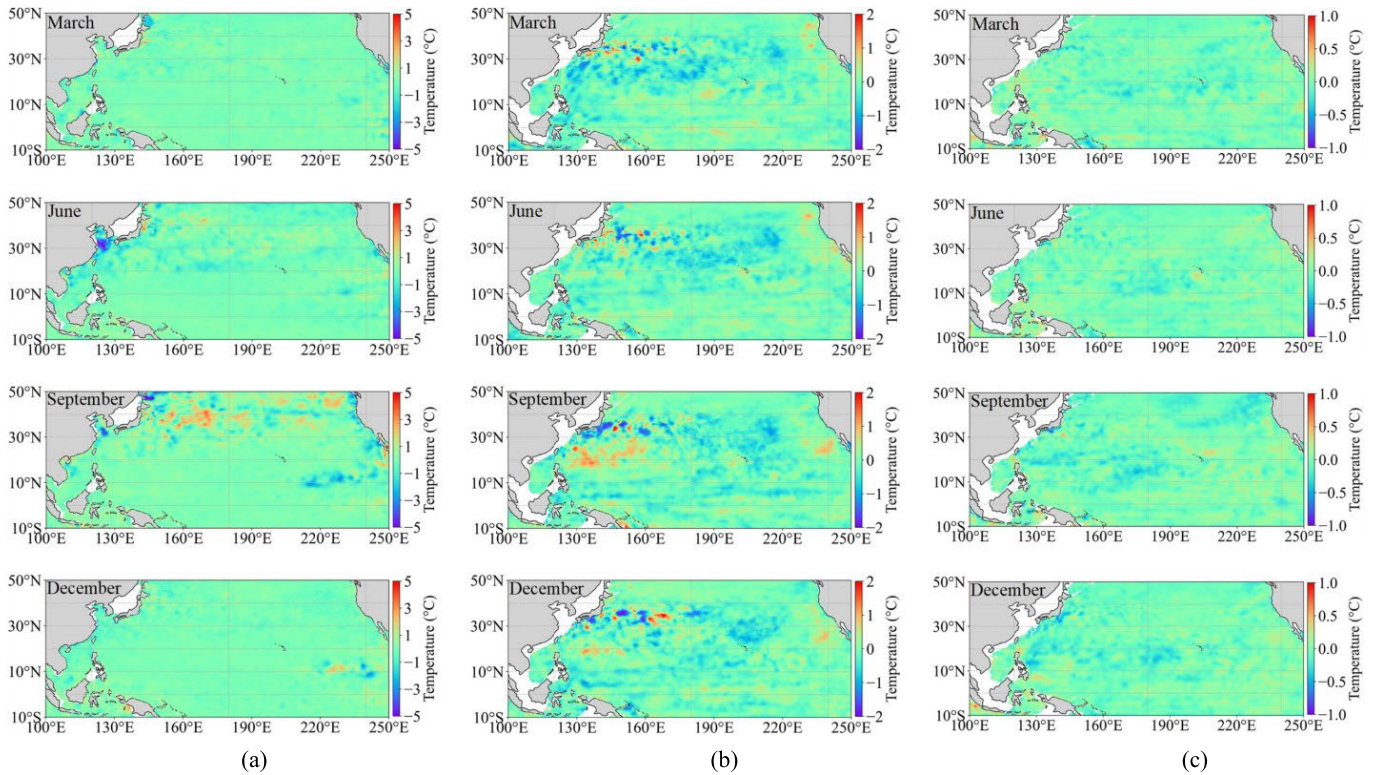


Fig. 10. Difference between CSSP-ConvLSTM reconstructed temperatures and ARMOR3D temperatures at different depths in different seasons: (a) 30, (b) 500, and (c) 1000 m.

the overall results still accurately capture the spatiotemporal distribution of seawater temperature. These results demonstrate that the CSSP-ConvLSTM model not only adapts well in

the shallow layers but also maintains high stability in deeper layers, providing an effective approach for reconstructing subsurface temperature in complex environments.

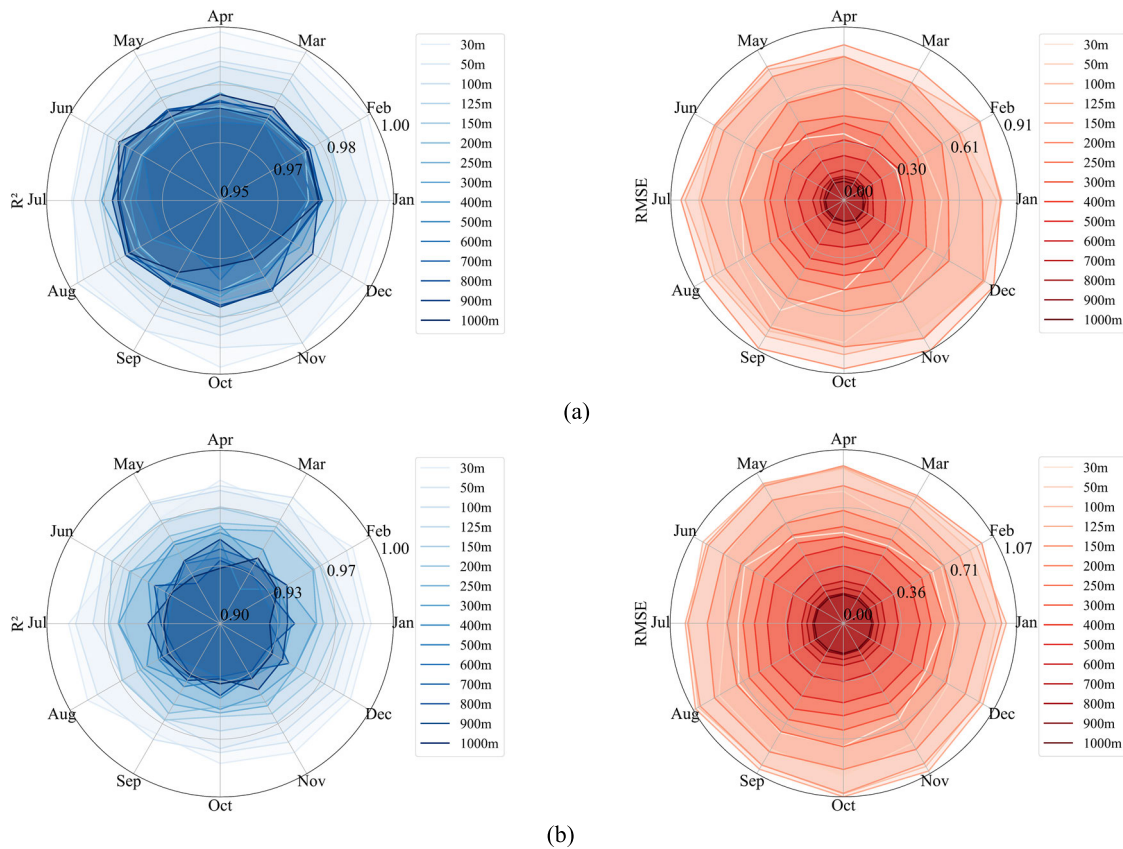


Fig. 11. R^2 and RMSE of CSSP-ConvLSTM reconstructed temperatures at different depth layers and different months. Monthly metrics are calculated over all spatial grid points at each depth. Evaluation of CSSP-ConvLSTM reconstruction results using (a) ARMOR3D ocean temperature and (b) GLORYS12V1 ocean temperature.

Additional inspection of the June and September results in Figs. 9 and 10 shows that reconstruction errors are moderately larger in dynamically active regions, such as the Kuroshio Extension and the North Equatorial Current, than in relatively quiescent areas. This increase is most evident at 30, 150, and 500 m depths, where strong currents and energetic mesoscale eddies enhance spatiotemporal heterogeneity, thereby increasing the difficulty of inferring subsurface thermal structures from surface observations. In contrast, regions with weaker dynamical activity exhibit smoother vertical structures, allowing the model to learn more stable statistical mappings and achieve higher reconstruction accuracy. Although error levels vary among different dynamical regimes, no persistent systematic bias is observed, indicating that CSSP-ConvLSTM maintains robust performance under complex oceanic conditions.

To further verify the spatiotemporal generalization of the CSSP-ConvLSTM, systematic assessments are conducted for 2022 using both ARMOR3D and GLORYS12V1 temperature products, as shown in Fig. 11. The ARMOR3D-based evaluation [see Fig. 11(a)] shows consistently high agreement throughout the year, with R^2 values exceeding 0.97 and RMSE remaining below 0.9 °C in most months, indicating strong reconstruction accuracy and stability. Although the R^2 value slightly decreases in February and October, and the RMSE is relatively high in January and February, overall accuracy remains high, and the reconstruction of temperature

evolution does not compromise the reliability. Independent evaluation using monthly GLORYS12V1 data interpolated to the 0.25° grid [see Fig. 11(b)] shows a modest reduction in accuracy, reflecting differences in data sources, assimilation strategies, and vertical representations between GLORYS12V1 and ARMOR3D. Nevertheless, CSSP-ConvLSTM maintains R^2 values above 0.91 and RMSE below 1.07 °C for most months, demonstrating satisfactory generalization capability across datasets. Overall, CSSP-ConvLSTM demonstrates strong spatiotemporal generalization and robustness in both stable and transitional seasons, enabling reliable reconstruction of subsurface temperature fields across all depths throughout the year.

C. Vertical Distribution Characteristics

The vertical distribution of CSSP-ConvLSTM reconstructions is evaluated by comparing reconstructed temperatures with ARMOR3D temperatures along meridional sections at longitudes 100°E , 130°E , 160°E , 190°E , 220°E , and 250°E and latitudinal sections at latitudes -10°N , 5°N , 20°N , 35°N , and 50°N for representative months of each season (March, June, September, and December) in 2022. Reconstructed temperatures, ARMOR3D temperatures, and their differences for different meridional and latitudinal profiles are shown in Figs. 12 and 13.

It is apparent from Figs. 12(a) and 13(a), the vertical structure of the temperature field is well captured by the

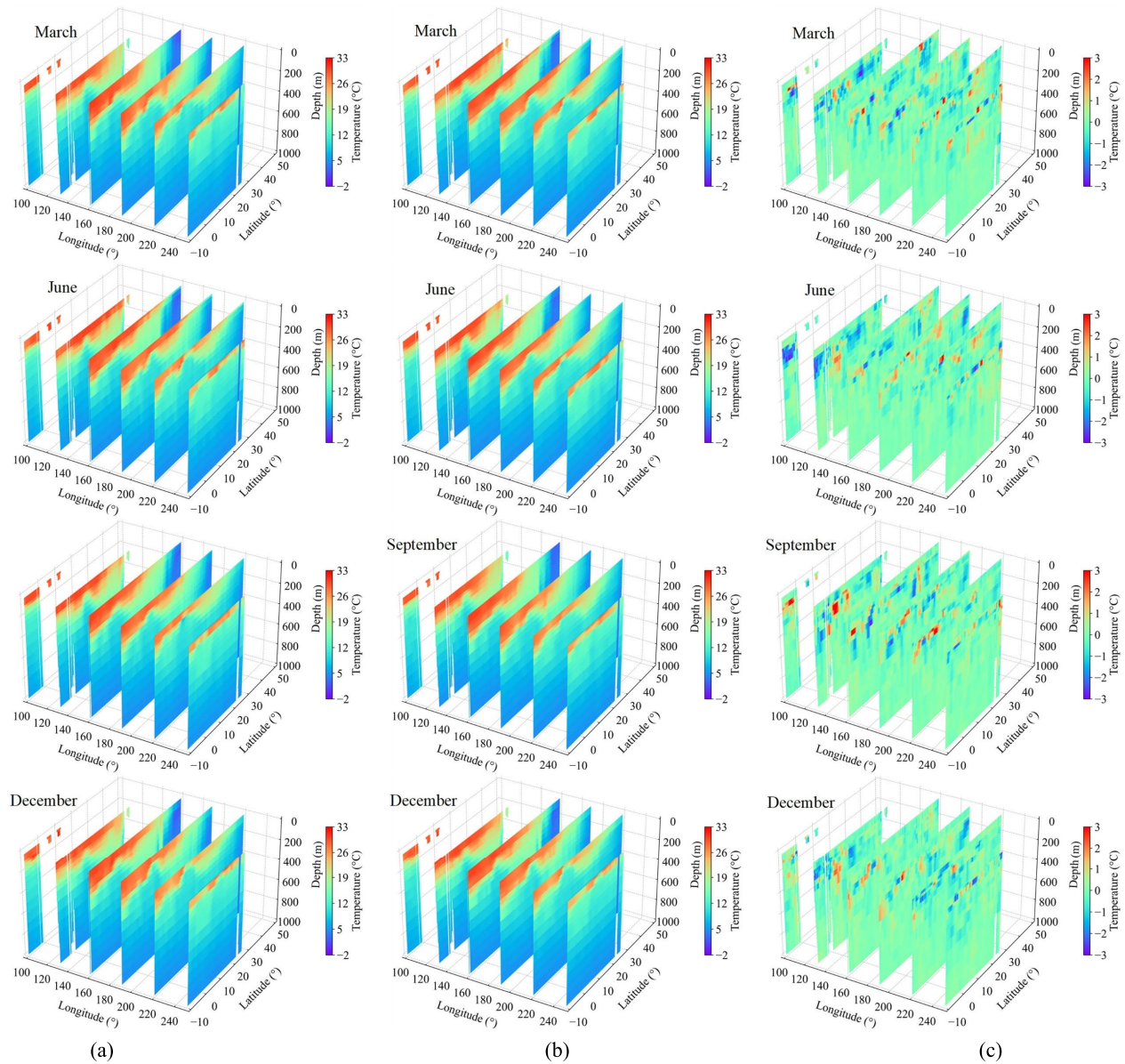


Fig. 12. Comparison of (a) CSSP-ConvLSTM reconstructed temperatures and (b) ARMOR3D temperatures at (c) different longitudinal profiles.

CSSP-ConvLSTM model across months and locations. Thermocline signatures are clearly delineated, and the overall distribution closely aligns with ARMOR3D. Figs. 12(b) and 13(b) show the cross-sectional distribution of ARMOR3D temperatures, which also reflects the main vertical variation characteristics.

However, differences in fine-scale temperature gradients are evident between ARMOR3D and the model. Figs. 12(c) and 13(c) illustrate the differences between reconstructed temperatures and ARMOR3D. Across most sections, the discrepancies are small, indicating a high degree of consistency with the ARMOR3D. In dynamically energetic regions (e.g., 0°N – 30°N and 120°E – 180°E), localized reconstruction errors can reach 2°C – 3°C within the 100–300-m layer. These error hotspots are consistent with the high-error regions identified in Figs. 9 and 10 in terms of both horizontal location and vertical structure, indicating that the spatial distribution of model errors

is closely linked to the underlying dynamical background. The increased errors, therefore, reflect the enhanced complexity of subsurface thermal structures in regions of intensified currents and eddy activity, rather than being determined solely by the relatively weak variability of the deep temperature field.

Taken together, the cross-month and cross-sectional comparisons indicate the strong effectiveness and reliability of the CSSP-ConvLSTM in reconstructing 3-D subsurface temperature. Although local errors persist in shallow regions, the real temperature distributions are reproduced accurately by the model overall.

D. Model Reconstruction Evaluation Based on En4 Data

The model reconstructions at different months and depths are systematically evaluated using the 2022 EN4.2.2 data. The EN4 data are linearly interpolated to the same 15 depths as the reconstruction results, and the reconstructions at the nearest

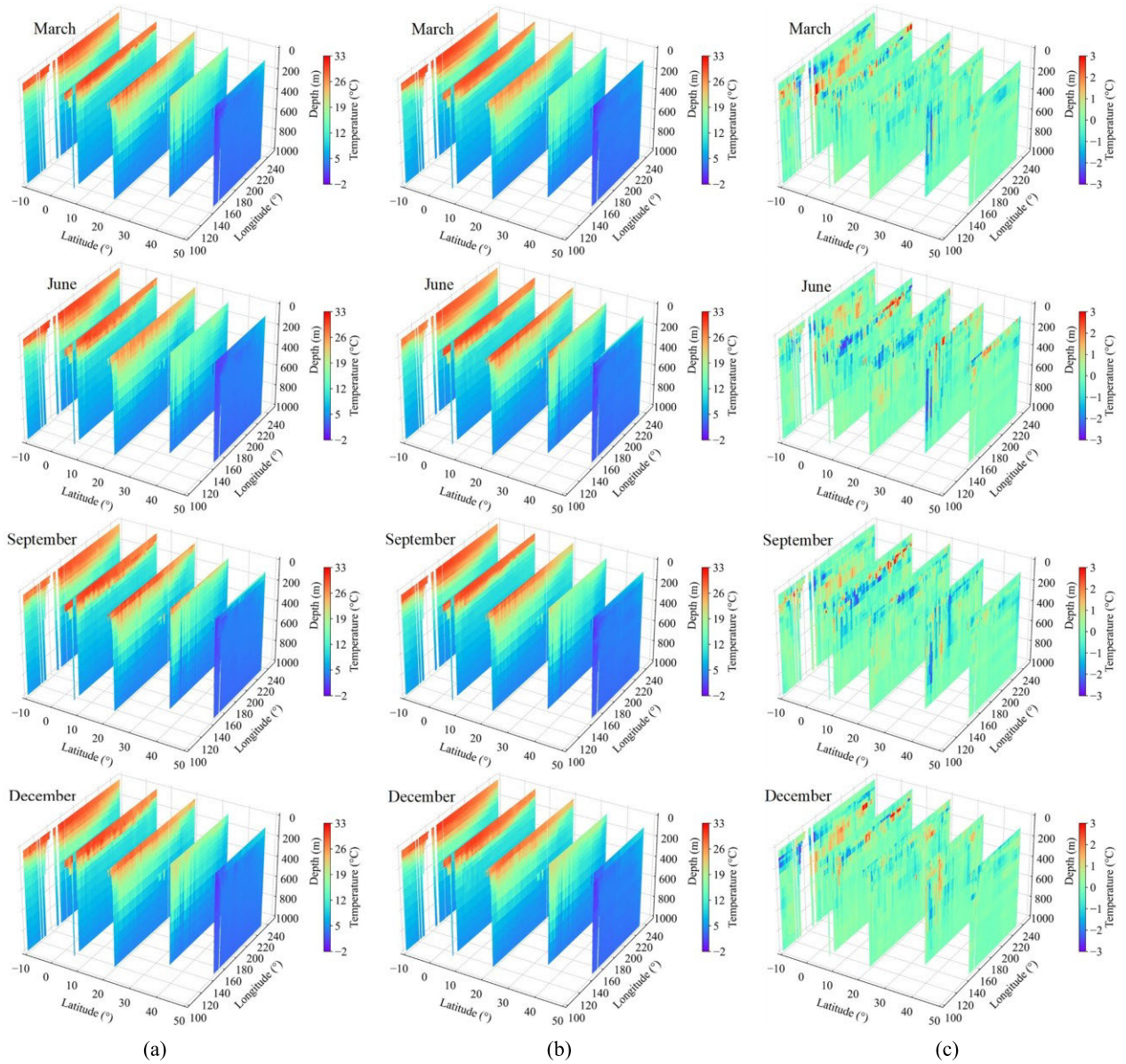


Fig. 13. Comparison of (a) CSSP-ConvLSTM reconstructed temperatures and (b) ARMOR3D temperatures at (c) different latitudinal profiles.

grid points within each layer are matched to the EN4 profiles. Heatmaps of R^2 and RMSE derived from comparisons between CSSP-ConvLSTM reconstructed temperatures, ARMOR3D temperatures, and EN4 temperature profiles are shown in Fig. 14. As shown in Fig. 14, the CSSP-ConvLSTM reconstructions exhibit high consistency across all months and depths throughout the year. The RMSE of the reconstructions compared with EN4 is generally lower than 0.6 °C in the thermocline and deep layers (100–1000 m). Although the shallow layer (30–100 m) is affected by seasonal fluctuations and surface forcing, the error is slightly higher but remains below 0.9 °C. The annual mean RMSE is 0.5781 °C. The R^2 distribution further verifies the robustness of the model, with an annual mean of 0.9513, indicating good agreement with the EN4 data. In comparison, the average R^2 and RMSE of ARMOR3D are 0.9421 °C and 0.6162 °C, respectively, reflecting slightly lower overall accuracy. In particular, the performance is significantly worse than that of

CSSP-ConvLSTM, especially in the shallow layers and during certain months. In summary, the CSSP-ConvLSTM model demonstrates marked advantages in capturing the complex structure of the thermocline and exhibits excellent reliability, temporal generalization, and robustness under varying temporal and depth conditions.

E. SHAP Analysis

To assess model interpretability, the importance of CSSP-ConvLSTM input variables is quantified using SHapley additive exPlanations (SHAP) [53]. SHAP values for input variables are computed at representative depths (30, 500, and 1000 m) to evaluate their relative contributions to subsurface temperature reconstruction, and the results are shown in Fig. 15. Fig. 15 indicates marked depth-dependent differences in variable importance. At 30 m, SST dominates the feature attribution, indicating that near-surface temperature variability

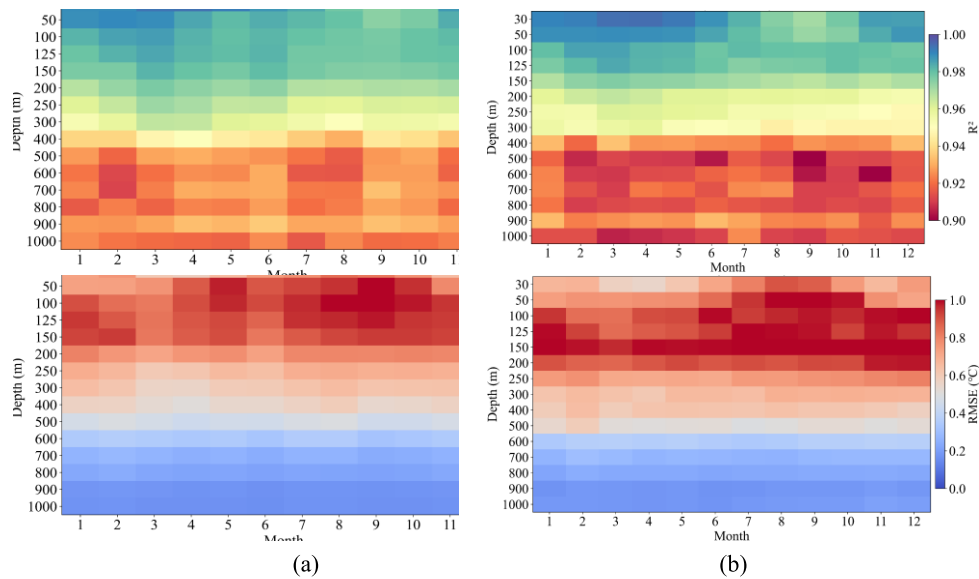


Fig. 14. Heatmap of RMSE and R^2 between (a) CSSP-ConvLSTM and EN4 and (b) ARMOR3D and EN4 data in 2022. This validation primarily assesses agreement at monthly and spatially averaged scales, while fine-scale structures may be smoothed in EN4.

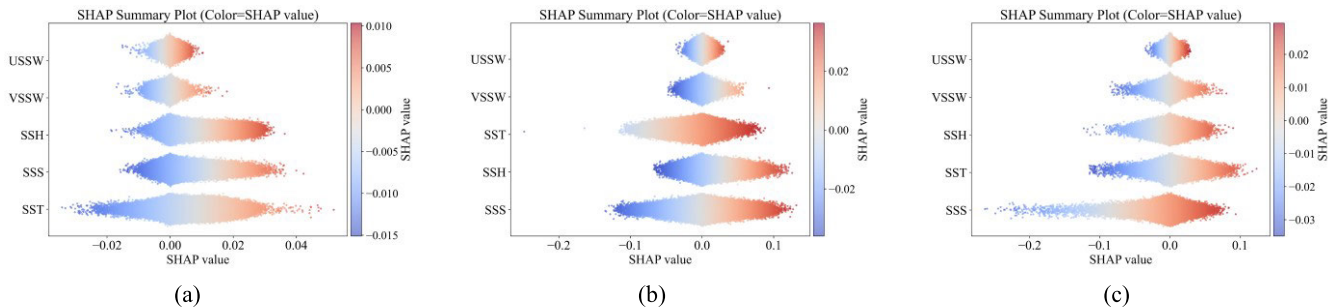


Fig. 15. SHAP summary plots illustrating the relative contributions of input variables in the CSSP-ConvLSTM model at selected depth layers. SHAP values along the horizontal axis indicate the direction and magnitude of each variable's influence on temperature reconstruction. Input variables are ranked by importance on the vertical axis. Each point represents a spatiotemporal sample, with colors denoting variable values (red: high; blue: low) (a) 30, (b) 500, and (c) 1000 m.

is primarily controlled by air–sea heat fluxes and mixed-layer processes. This result is consistent with the strong coupling between mixed-layer temperature and surface thermal conditions. At 500 m, the relative importance of SSS and SSH increases markedly. In particular, SSH, which reflects large-scale dynamical structures and isopycnal displacements, plays an important role in modulating the mid-depth thermal distribution, while SSS indirectly influences temperature through its control of the density structure and stratification stability. At 1000 m, SSS becomes the dominant contributor, suggesting that deep-layer temperature variability is primarily governed by water-mass properties and the density-controlled thermohaline structure. Across all examined depths, the USSW and VSSW remain relatively small, indicating that wind forcing primarily affects subsurface temperature through indirect dynamical pathways rather than direct thermal control. Overall, the SHAP analysis demonstrates that the key input variables relied upon by CSSP-ConvLSTM at different depths are physically consistent with established thermohaline structures and ocean dynamical processes, thereby supporting the physical plausibility and interpretability of the reconstruction results.

V. CONCLUSION

To address the persistent challenge of limited thermocline accuracy in 3-D ocean temperature reconstruction, a subsurface temperature reconstruction model based on multisource sea surface remote sensing is developed, termed CSSP-ConvLSTM. Multisource surface observations, along with 3-D data products, are utilized to achieve high-precision reconstruction of the subsurface temperature field over the depth range of 30–1000 m. By incorporating a spatial dual-branch architecture and temporal enhancement mechanisms, CSSP-ConvLSTM effectively captures both local spatial details and nonlocal dependencies, leading to substantial improvements in the reconstruction of thermocline and deep-layer thermal structures. For spatial feature extraction, a dual-branch configuration combining CNN and Swin Transformer, augmented with the CBAM module, provides complementary local perception and hierarchical global representation. This design proves particularly effective within the thermocline (100–600 m), where strong vertical gradients and complex spatial variability are present. For temporal feature extraction, ConvLSTM integrated with TCA and SE modules enhances

the representation of temporal evolution while suppressing redundant information, resulting in robust temporal generalization and stability under varying seasonal forcing.

Reconstruction experiments conducted in the northwestern Pacific demonstrate that CSSP-ConvLSTM consistently outperforms CNN-ConvLSTM, Swin-ConvLSTM, and other baseline architectures. For the year 2022, the model achieves a depth-averaged R^2 of 0.9811 and an RMSE of 0.4557 °C. Pronounced error reductions are observed in both the thermocline (100–600 m) and deep layers (800–1000 m), highlighting the enhanced capability of the model in capturing complex 3-D thermal structures.

Independent evaluations further confirm the robustness and generalization capability of the proposed approach. Validation against ARMOR3D shows high consistency across different months and depth levels, with only modest error increases in the surface mixed layer during strong seasonal disturbances. Thermocline reconstructions remain stable, and deep-layer errors are largely confined within ± 0.5 °C. Evaluation using GLORYS reanalysis data reveals a moderate reduction in accuracy relative to ARMOR3D, consistent with known differences in dynamical frameworks, assimilation strategies, and vertical representations. Nevertheless, R^2 values generally remain above 0.91 and RMSE below 1.1 °C across most months and depths, indicating satisfactory cross-product transferability. Additional validation against EN4 in situ profiles further corroborates the reconstruction skill, yielding an average R^2 of 0.9513 and RMSE of 0.5781 °C, which are overall slightly superior to those of ARMOR3D, with particularly evident improvements in the thermocline.

The statistical reliability of the reconstruction is further quantified through mean error and depthwise 95% CIs based on spatial samples from 2010 to 2022. Mean errors remain close to zero and are primarily within ± 0.1 °C, indicating the absence of pronounced systematic bias. CIs are generally narrow, reflecting stable estimation of spatially averaged reconstruction errors, while a modest widening is observed in the 50–300 m depth range due to strong stratification and enhanced dynamical complexity. Consistent with the spatial error distributions, dynamically energetic regions such as the Kuroshio Extension and the North Equatorial Current exhibit moderately larger shallow-to-mid-depth errors than relatively quiescent regions, without persistent bias, demonstrating stable model performance under heterogeneous ocean dynamical conditions.

Model interpretability analysis based on SHAP reveals pronounced depth-dependent variations in input-variable importance. SST dominates near the surface (30 m), SSS and SSH become increasingly influential at mid-depth (500 m), and SSS emerges as the primary contributor at 1000 m. These depth-dependent attribution patterns are consistent with established physical oceanographic understanding, supporting both the interpretability and physical plausibility of CSSP-ConvLSTM and offering additional insights into subsurface thermal dynamics.

In summary, by combining a spatial dual-branch structure with temporal enhancement, CSSP-ConvLSTM enables high-precision reconstruction of subsurface temperature fields,

with particularly strong performance in the thermocline and deep ocean. The model thus provides a novel approach to obtain high-resolution 3-D subsurface temperature fields and establishes a methodological basis for improved understanding of ocean dynamics and associated climate impacts. Future work may incorporate additional satellite observations and explicit physical constraints to extend its applicability for joint reconstruction of 3-D thermohaline and dynamical fields.

ACKNOWLEDGMENT

The authors are particularly grateful to the Copernicus Marine Service for the distribution of SSH, SSS, ARMOR3D, and GLORYS12V1 data; UKMO for the distribution of SST data; CCMP for SSW data; and Met Office for EN4 data.

REFERENCES

- [1] S.-P. Xie and Y. Kosaka, "What caused the global surface warming hiatus of 1998–2013?," *Current Climate Change Rep.*, vol. 3, no. 2, pp. 128–140, Jun. 2017, doi: [10.1007/s40641-017-0063-0](https://doi.org/10.1007/s40641-017-0063-0).
- [2] Y. Li et al., "Marine resource economy and strategy under the background of marine ecological civilization construction," *J. Nat. Resour.*, vol. 37, no. 4, pp. 829–849, Apr. 2022, doi: [10.31497/zrzyxb.20220407](https://doi.org/10.31497/zrzyxb.20220407).
- [3] L. D. Talley, G. L. Pickard, and W. J. Emery, "Physical properties of seawater," in *Descriptive Physical Oceanography*, 6th ed., New York, NY, USA: Academic, 2011, pp. 29–65.
- [4] National Oceanic and Atmospheric Administration (NOAA). (2024). *What is a Thermocline?*. Accessed: Jan. 2026. [Online]. Available: <https://oceanservice.noaa.gov/facts/thermocline.html>
- [5] J. Pedlosky, "A history of thermocline theory," in *Proc. J. Wang C. N. K. Mooers*, 2006, pp. 139–152.
- [6] C. Dong et al., "Prospect of artificial intelligence in oceanography," *J. Mar. Sci.*, vol. 42, no. 3, pp. 2–27, Mar. 2024. [Online]. Available: <https://www.sciengine.com/doi/10.3969/j.issn.1001-909X.2024.03.001>
- [7] N. Liu, H. Wang, T. Ling, and Z. Zu, "Review and prospect of global operational ocean forecasting," *Adv. Earth Sci.*, vol. 33, no. 2, pp. 131–140, Feb. 2018, doi: [10.11867/j.issn.1001-8166.2018.02.0131](https://doi.org/10.11867/j.issn.1001-8166.2018.02.0131).
- [8] D. Roemmich et al., "On the future of argo: A global, full-depth, multi-disciplinary array," *Frontiers Mar. Sci.*, vol. 6, Mar. 2019, Art. no. 439, doi: [10.3389/fmars.2019.00439](https://doi.org/10.3389/fmars.2019.00439).
- [9] M. M. Ali, D. Swain, and R. A. Weller, "Estimation of ocean subsurface thermal structure from surface parameters: A neural network approach," *Geophys. Res. Lett.*, vol. 31, no. 20, 2004, Art. no. L20308, doi: [10.1029/2004gl021192](https://doi.org/10.1029/2004gl021192).
- [10] J. Holloway and K. Mengersen, "Statistical machine learning methods and remote sensing for sustainable development goals: A review," *Remote Sens.*, vol. 10, no. 9, p. 1365, Aug. 2018, doi: [10.3390/rs10091365](https://doi.org/10.3390/rs10091365).
- [11] L. Cheng, J. Abraham, Z. Hausfather, and K. E. Trenberth, "How fast are the oceans warming?," *Science*, vol. 363, no. 6423, pp. 128–129, Jan. 2019, doi: [10.1126/science.aav7619](https://doi.org/10.1126/science.aav7619).
- [12] Y. Jeong, J. Hwang, J. Park, C. J. Jang, and Y.-H. Jo, "Reconstructed 3-D ocean temperature derived from remotely sensed sea surface measurements for mixed layer depth analysis," *Remote Sens.*, vol. 11, no. 24, p. 3018, Dec. 2019, doi: [10.3390/rs11243018](https://doi.org/10.3390/rs11243018).
- [13] O. M. Smedstad and J. J. O'Brien, "Variational data assimilation and parameter estimation in an equatorial Pacific Ocean model," *Prog. Oceanogr.*, vol. 26, no. 2, pp. 179–241, 1991, doi: [10.1016/0079-6611\(91\)90002-4](https://doi.org/10.1016/0079-6611(91)90002-4).
- [14] R. R. Rao and R. Sivakumar, "Seasonal variability of sea surface salinity and salt budget of the mixed layer of the north Indian Ocean," *J. Geophys. Res., Oceans*, vol. 108, no. C1, Jan. 2003, Art. no. 3009, doi: [10.1029/2001jc000907](https://doi.org/10.1029/2001jc000907).
- [15] V. Klemas and X.-H. Yan, "Subsurface and deeper ocean remote sensing from satellites: An overview and new results," *Prog. Oceanogr.*, vol. 122, pp. 1–9, Mar. 2014, doi: [10.1016/j.pocan.2013.11.010](https://doi.org/10.1016/j.pocan.2013.11.010).
- [16] J. Wang, G. R. Flierl, J. H. LaCasce, J. L. McClean, and A. Mahadevan, "Reconstructing the Ocean's interior from surface data," *J. Phys. Oceanogr.*, vol. 43, no. 8, pp. 1611–1626, Aug. 2013, doi: [10.1175/jpo-d-12-0204.1](https://doi.org/10.1175/jpo-d-12-0204.1).

- [17] L. Liu, S. Peng, J. Wang, and R. X. Huang, "Retrieving density and velocity fields of the ocean's interior from surface data," *J. Geophys. Res., Oceans*, vol. 119, no. 12, pp. 8512–8529, Dec. 2014, doi: [10.1002/2014jc010221](#).
- [18] J. Mu, J. Yang, C. Wang, and Y. Jia, "OSTSI-CBFNet: Ocean subsurface temperature and salinity inversion based on CNN-BiLSTM fusion network with attention mechanism," *J. Appl. Remote Sens.*, vol. 18, no. 3, Sep. 2024, Art. no. 038507, doi: [10.1117/1.jrs.18.038507](#).
- [19] G. Chen, B. Huang, X. Chen, L. Ge, M. Radenkovic, and Y. Ma, "Deep blue AI: A new bridge from data to knowledge for the ocean science," *Deep Sea Res. I, Oceanographic Res. Papers*, vol. 190, Dec. 2022, Art. no. 103886, doi: [10.1016/j.dsr.2022.103886](#).
- [20] T. Song et al., "A review of artificial intelligence in marine science," *Frontiers Earth Sci.*, vol. 11, Feb. 2023, Art. no. 1090185, doi: [10.3389/feart.2023.1090185](#).
- [21] H. Zheng, H. Bi, X. Cheng, and M. C. Benfield, "Editorial: Deep learning for marine science," *Frontiers Mar. Sci.*, vol. 11, May 2024, Art. no. 1407053, doi: [10.3389/fmars.2024.1407053](#).
- [22] Y. Lecun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proc. IEEE*, vol. 86, no. 11, pp. 2278–2324, Nov. 1998, doi: [10.1109/5.726791](#).
- [23] T. Bolton and L. Zanna, "Applications of deep learning to ocean data inference and subgrid parameterization," *J. Adv. Model. Earth Syst.*, vol. 11, no. 1, pp. 376–399, Jan. 2019, doi: [10.1029/2018ms001472](#).
- [24] H. Su, A. Wang, T. Zhang, T. Qin, X. Du, and X.-H. Yan, "Super-resolution of subsurface temperature field from remote sensing observations based on machine learning," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 102, Oct. 2021, Art. no. 102440, doi: [10.1016/j.jag.2021.102440](#).
- [25] K. Mao, C. Liu, S. Zhang, and F. Gao, "Reconstructing ocean subsurface temperature and salinity from sea surface information based on dual path convolutional neural networks," *J. Mar. Sci. Eng.*, vol. 11, no. 5, p. 1030, May 2023, doi: [10.3390/jmse11051030](#).
- [26] J. Qi, B. Xie, D. Li, J. Chi, B. Yin, and G. Sun, "Estimating thermohaline structures in the tropical Indian Ocean from surface parameters using an improved CNN model," *Frontiers Mar. Sci.*, vol. 10, Apr. 2023, Art. no. 1181182, doi: [10.3389/fmars.2023.1181182](#).
- [27] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 5998–6008.
- [28] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022, doi: [10.1109/ICCV48922.2021.00986](#).
- [29] Y. Yu, X. Si, C. Hu, and J. Zhang, "A review of recurrent neural networks: LSTM cells and network architectures," *Neural Comput.*, vol. 31, no. 7, pp. 1235–1270, Jul. 2019, doi: [10.1162/neco_a_01199](#).
- [30] L. Wei, L. Guan, L. Qu, and D. Guo, "Prediction of sea surface temperature in the China seas based on long short-term memory neural networks," *Remote Sens.*, vol. 12, no. 17, p. 2697, Aug. 2020, doi: [10.3390/rs12172697](#).
- [31] X. Jia, Q. Ji, L. Han, Y. Liu, G. Han, and X. Lin, "Prediction of sea surface temperature in the East China Sea based on LSTM neural network," *Remote Sens.*, vol. 14, no. 14, p. 3300, Jul. 2022, doi: [10.3390/rs14143300](#).
- [32] X. Shi, Z. Chen, H. Wang, D. Yeung, W. K. Wong, and W. Woo, "Convolutional LSTM network: A machine learning approach for precipitation nowcasting," in *Proc. Adv. Neural Inf. Process. Syst.*, 2015, pp. 802–810.
- [33] K. Zhang, X. Geng, and X.-H. Yan, "Prediction of 3-D ocean temperature by multilayer convolutional LSTM," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 8, pp. 1303–1307, Aug. 2020, doi: [10.1109/LGRS.2019.2947170](#).
- [34] Y. Zhang, Y. Liu, Y. Kong, and P. Hu, "An improved method for retrieving subsurface temperature using the ConvLSTM model in the western Pacific Ocean," *J. Mar. Sci. Eng.*, vol. 12, no. 4, p. 620, Apr. 2024, doi: [10.3390/jmse12040620](#).
- [35] E. Yankovsky, L. Zanna, and K. S. Smith, "Influences of mesoscale ocean eddies on flow vertical structure in a resolution-based model hierarchy," *J. Adv. Model. Earth Syst.*, vol. 14, no. 11, Nov. 2022, Art. no. e2022MS003203, doi: [10.1029/2022ms003203](#).
- [36] E. Romero, L. Tenorio-Fernandez, E. Portela, J. Montes-Aréchiga, and L. Sánchez-Velasco, "Improving the thermocline calculation over the global ocean," *Ocean Sci.*, vol. 19, no. 3, pp. 887–901, Jun. 2023, doi: [10.5194/os-19-887-2023](#).
- [37] C. J. Donlon, M. Martin, J. Stark, J. Roberts-Jones, E. Fiedler, and W. Wimmer, "The operational sea surface temperature and sea ice analysis (OSTIA) system," *Remote Sens. Environ.*, vol. 116, pp. 140–158, Jan. 2012, doi: [10.1016/j.rse.2010.10.017](#).
- [38] G. Taburet et al., "DUACS DT2018: 25 years of reprocessed sea level altimetry products," *Ocean Sci.*, vol. 15, no. 5, pp. 1207–1224, Sep. 2019, doi: [10.5194/os-15-1207-2019](#).
- [39] M. Segal-Rozenhaimer, A. Li, K. Das, and V. Chirayath, "Cloud detection algorithm for multi-modal satellite imagery using convolutional neural-networks (CNN)," *Remote Sens. Environ.*, vol. 237, Feb. 2020, Art. no. 111446, doi: [10.1016/j.rse.2019.111446](#).
- [40] R. Atlas et al., "A cross-calibrated, multiplatform ocean surface wind velocity product for meteorological and oceanographic applications," *Bull. Amer. Meteorological Soc.*, vol. 92, no. 2, pp. 157–174, Feb. 2011, doi: [10.1175/2010bams2946.1](#).
- [41] S. Guinehut, A.-L. Dhomp, G. Larnicol, and P.-Y. Le Traon, "High resolution 3-D temperature and salinity fields derived from in situ and satellite observations," *Ocean Sci.*, vol. 8, no. 5, pp. 845–857, Oct. 2012, doi: [10.5194/os-8-845-2012](#).
- [42] *The Copernicus Global 1/12° Oceanic and Sea Ice Physical Reanalysis (GLORYS12V1)*, Mercator Ocean International (Copernicus Marine Service, CMEMS), Toulouse, France, 2021.
- [43] S. A. Good, M. J. Martin, and N. A. Rayner, "EN4: Quality controlled ocean temperature and salinity profiles and monthly objective analyses with uncertainty estimates," *J. Geophys. Res., Oceans*, vol. 118, no. 12, pp. 6704–6716, Dec. 2013, doi: [10.1002/2013jc009067](#).
- [44] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis.*, Sep. 2018, pp. 3–19, doi: [10.1007/978-3-030-01234-2_1](#).
- [45] M. Lin, Q. Chen, and S. Yan, "Network in network," 2013, *arXiv:1312.4400*.
- [46] L. Lin, J. Wu, S. Fu, S. Zhang, C. Tong, and L. Zu, "Channel attention & temporal attention based temporal convolutional network: A dual attention framework for remaining useful life prediction of the aircraft engines," *Adv. Eng. Informat.*, vol. 60, Apr. 2024, Art. no. 102372, doi: [10.1016/j.aei.2024.102372](#).
- [47] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141, doi: [10.1109/CVPR.2018.00745](#).
- [48] H. Xie, C. Dong, and Q. Xu, "Dual U-vision-transformer for reconstructing the three-dimensional eddy-resolving oceanic physical parameters from satellite observations," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 136, Feb. 2025, Art. no. 104382, doi: [10.1016/j.jag.2025.104382](#).
- [49] J. Neyman, "Outline of a theory of statistical estimation based on the classical theory of probability," *Phil. Trans. Roy. Soc. London. Ser. A, Math. Phys. Sci.*, vol. 236, no. 767, pp. 333–380, Aug. 1937, doi: [10.1098/rsta.1937.0005](#).
- [50] Intergovernmental Panel on Climate Change (IPCC), *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I To the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*, V. Masson-Delmotte et al., Eds., Cambridge, U.K.: Cambridge Univ. Press, 2021, p. 2391, doi: [10.1017/9781009157896](#).
- [51] National Centers for Environmental Information (NCEI). (2023). *Annual 2022 Tropical Cyclones Report*. [Online]. Available: <https://www.ncei.noaa.gov>
- [52] A. Wada, "The processes of SST cooling by typhoon passage and case study of typhoon rex with a mixed layer ocean model," *Papers Meteorol. Geophysics*, vol. 52, no. 2, pp. 31–66, 2002, doi: [10.2467/mripapers.52.31](#).
- [53] S. Lundberg and S. Lee, "A unified approach to interpreting model predictions," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 4765–4774.



Jie Sun received the M.E. degree in surveying engineering from Shandong University of Science and Technology, Qingdao, China, in 2020. She is currently pursuing the Ph.D. degree with the College of Intelligent Systems Science and Engineering, Harbin Engineering University, Harbin, China.

Her research interests mainly include ocean remote sensing, deep learning, and ocean subsurface temperature and salinity inversion.



Jungang Yang (Member, IEEE) received the M.S. degree in mathematics from Inner Mongolia University, Hohhot, China, in 2004, and the Ph.D. degree in physical oceanography from the Institute of Oceanology, Chinese Academy of Sciences, Qingdao, China, in 2007.

In 2007, he joined the Laboratory of Marine Physics and Remote Sensing, First Institute of Oceanography, Ministry of Natural Resources, Qingdao. His research interests include marine dynamic environment remote sensing, satellite altimeter data processing and applications, and marine underwater dynamic environment intelligent detection.



Ole Baltazar Andersen received the M.E. degree in geophysics from the University of Aarhus, Aarhus, Denmark, in 1992, and the Ph.D. degree in geophysics from the University of Copenhagen, København, Denmark, in 1996.

He is with DTU Space, Technical University of Denmark, Kongens Lyngby, Denmark. He was on the science working teams for the NASA–CNES TOPEX/Poseidon, JASON-1, and GRACE missions. His research interests include satellite altimetry, ocean tide modeling, and geodetic and oceanographic applications of Earth observation data.

Dr. Andersen is an Assistant Secretary-General of the International Association of Geodesy.



Wei Cui received the M.E. degree in physical oceanography from the First Institute of Oceanography, Qingdao, China, in 2016, and the Ph.D. degree in physical oceanography from the Ocean University of China, Qingdao, in 2019.

He is currently an Associate Research Fellow at the First Institute of Oceanography, Ministry of Natural Resources (MNR). His research interests include satellite altimeter data analysis and oceanic eddy identification.



Jiawei Xun received the B.E. degree in communication engineering from China University of Petroleum (East China), Qingdao, China, in 2023, where he is currently pursuing the M.E. degree with the College of Oceanography and Space Informatics.

His research interests mainly include deep learning and ocean subsurface temperature and salinity inversion.